Reconstruction of the Evolutionary Landscape of Biological Processes Involved in the Early Stages of the Metastatic Cascade

Evolutionary Origins of Metastasis

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Abstract

Metastasis is not a *de novo* functional module innovation but rather the pathological redeployment of deeply conserved biological programs. Here, we reconstruct the evolutionary landscape of biological functions involved in the early stages of the metastatic cascade, focusing on cell adhesion, extracellular matrix organization (ECM), regulation of metalloproteinase activity, cell junction organization, epithelial–mesenchymal transition (EMT), and cellular extravasation. Using phyletic pattern reconstruction of 668 orthologs across 396 Cluster of Orthologous Groups with the GeneBridge algorithm, we identified three human Last Common Ancestor (LCA) evolutionary nodes of major enrichment: Human–Discoba, Human–Ctenophora, Human-Porifera, and Human–Actinopterygii. ECM organization was significantly enriched in Discoba, EMT in Ctenophora, cell junction organization in Porifera, and extravasation in Actinopterygii, while adhesion and metalloproteinase regulation showed dispersed roots across clades. Our findings demonstrate that metastasis-related modules were progressively assembled across evolutionary history, with pleiotropic ancestral orthologs—initially essential for cellular homeostasis—co-opted for tumor dissemination. This evolutionary perspective refines our understanding of metastasis as a reconfiguration of ancient biological processes rather than a strictly cancer-specific innovation.

1. Introduction

Metastasis, a hallmark of aggressive cancers, represents a complex, multiphasic process in which primary tumor cells disseminate to distant tissues, forming secondary tumor *foci*. This biological cascade involves distinct cellular events: invasion of adjacent tissues, intravasation into the circulatory system, survival in circulation, extravasation, and colonization of new organs (Pachmayr et al., 2017; Welch & Hurst, 2019; Fares et al., 2020).

Many processes central to metastasis, including cell adhesion and epithelial–mesenchymal transition (EMT), have orthologs in evolutionarily distant lineages, extending even to unicellular organisms (Olson, 2006; Castro et al., 2008, 2009; Robert, 2011; Chen et al., 2015; Viscardi et al., 2021; Feuermann et al., 2025). This conservation suggests that the metastatic potential of cancer cells is not an exclusively pathological innovation but rather a consequence of genetic programs that were established early in eukaryotic evolution (Chen et al., 2015). Identifying the origin of these orthologs can offer insights into how metastasis-related biological processes emerged as a biological phenomenon.

To elucidate the evolutionary origins of cancer, Chen and collaborators (2015) traced the emergence of orthologous human protein-coding genes across 13 major clades, revealing a significant overrepresentation of cancer-driver genes at the origin of multicellularity. However, such deep approaches are computationally intensive and often challenging to scale. Consequently, alternative methods leveraging the phyletic patterns of orthologs have proven effective for reconstructing the history of other complex systems. A key example is the use of tools like GeneBridge with the STRING dataset to reconstruct the orthologs network across five major neurotransmitter systems (Viscardi et al., 2021; Campos et al., 2024).

Therefore, this study aims to map the evolutionary trajectory of key biological processes involved in the early stages of the metastatic cascade. Specifically, we focus on fundamental metastatic processes: cell adhesion, extracellular matrix (ECM) organization, EMT, regulation of metalloprotease activity, extravasation, and cell junction organization. We aim to uncover evolutionary orthologs' phyletic patterns and functional shifts that may have contributed to the emergence of metastasis-related traits. This evolutionary perspective not only deepens our understanding of the genetic foundations of metastasis but also establishes a framework for future studies investigating other stages of metastatic progression.

2. Materials and Methods

This study investigated the evolution of the early stages of the metastatic cascade biological processes, as detailed in Figure 1. Biological processes were selected from the Gene Ontology (GO) database using AmiGO 2, focusing on experimentally supported terms. Associated genes were then identified for evolutionary analyses, employing computational tools to determine the most probable evolutionary root and explore patterns across clades.

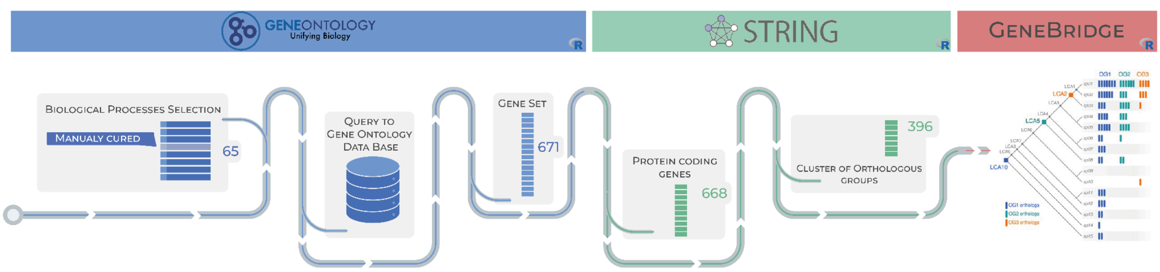


Figure 1. Methodology workflow overview. Biological processes related to metastasis were selected from the Gene Ontology database. Using the GO API, a list of genes associated with each process was retrieved. The STRING API was then used to map these genes to STRING IDs, followed by filtering to retain only genes annotated with at least one COG category. Subsequent analyses included protein-protein interaction network reconstruction, gene rooting using the Last Common Ancestor (LCA) approach, and gene abundance estimation across taxa.

2.1. Early metastasis biological process selection

Biological pathways associated with the early stages of the metastatic cascade, including local invasion, intravasation, survival in the circulatory system, and extravasation (Fidler et al., 2003; Valastyan et al., 2011), were selected using AmiGO 2 (v. 2.5.17) (Carbon et al., 2009). Only terms (including child terms) with experimental evidence directly related to key biological processes implicated in metastasis were included, resulting in a set of 65 biological processes (Table 1). For the purpose of this study, these biological processes will be referred to by their parent terms, namely: cell adhesion, ECM organization, regulation of metallopeptidase activity, cell junction organization, cellular extravasation, and EMT. The 65 biological processes were used to query the AmiGO 2 database, which returned 671 unique genes in total. The final gene set was then used in downstream analyses.

2.2. Evolutionary rooting analysis

To identify the evolutionary history of the early metastasis biological processes, an evolutionary rooting analysis was performed using GeneBridge Bioconductor R package (v. 0.99.2) (Campos et al., 2024). GeneBridge employs the Bridge algorithm to determine the Last Common Ancestor (LCA) for genes within specific Orthologous Groups (OGs), thereby inferring their evolutionary origin on a given phylogenetic tree. Of the 671 genes previously identified as relevant to metastasis, 668 were protein-coding and annotated with at least one Cluster of Orthologous Genes (COG) (Tatusov et. al., 2003) identifier and were thus included in our analysis. We first used GeneBridge to determine the most likely evolutionary root of their corresponding OGs. Subsequently, we conducted a two-step statistical analysis: (1) we applied a modified z-test to identify clades with a significant outlier count of orthologs emergence events from our set of 668 protein-coding genes, and (2) for the orthologs originating in each clades, we performed a hypergeometric test to assess the enrichment of each of the 65 biological processes. This analysis was based on the 476-eukaryote phylogenetic tree from STRING v.11 (Szklarczyk et. al, 2019), using pre-processed COG annotations sourced from the geneplast.data (v0.99.9) Bioconductor R package (Campos et al., 2020).

2.3. Computational Analysis Environment

All analyses were performed in the R environment (v4.4.2) using the High-Performance Computing Center at UFRN (NPAD/UFRN). Each analysis step was executed separately, and the complete repository can be accessed at github.com/dalmolingroup/metastasisevolution.

2.4. Statistical tests

We employed two statistical approaches to investigate the evolutionary origins of genes involved in the early stages of metastasis. First, to identify clades showing a significant increase in ortholog emergence, we used a modified z-score based on the Median Absolute Deviation (MAD-z-score). This measure is more robust than the conventional z-score because ortholog emergence values vary widely across the phylogenetic tree. Second, we applied a hypergeometric test to assess which of the 65 biological processes were statistically enriched within these clades. Parameters for the test included the total number of orthologs, the number of orthologs in a given clade, the number of orthologs associated with a biological process, and the overlap between clade-specific orthologs and process-associated orthologs. P-values were computed using the *phyper* function in R (v4.4.2) and adjusted for multiple testing with the Benjamini–Hochberg (BH) method.

3. Results

3.1. Evolutionary Network of Biological Processes Involved in the Initial Stages of the Metastatic Cascade

Our analysis began with the identification of genes associated with six biological processes fundamental to the early metastatic cascade: cell adhesion, ECM organization, regulation of metallopeptidase activity, cell junction organization, cellular extravasation, and EMT. By leveraging COG annotations from the STRING database, this initial set was filtered to a final dataset comprising. Our analysis began with the identification of genes associated with six biological processes fundamental to the early metastatic cascade: cell adhesion, ECM organization, regulation of metallopeptidase activity, cell junction organization, cellular extravasation, and EMT. By leveraging COG annotations from the STRING database, this initial set was filtered to a final dataset comprising 668 genes distributed across 396 COGs. This set of genes had the vast majority participating in multiple processes rather than being exclusive to a single one (Figure 2). Subsequently, we performed an evolutionary rooting analysis on each of these 396 OGs using the Genebridge R package. This analysis aimed to identify the most ancient, vertically inherited archetype for each OG throughout evolution.668 genes distributed across 396 COGs. This set of genes had the vast majority participating in multiple processes rather than being exclusive to a single one (Figure 2). Subsequently, we performed an evolutionary rooting analysis on each of these 396 OGs using the Genebridge R package. This analysis aimed to identify the most ancient, vertically inherited archetype for each OG throughout evolution.

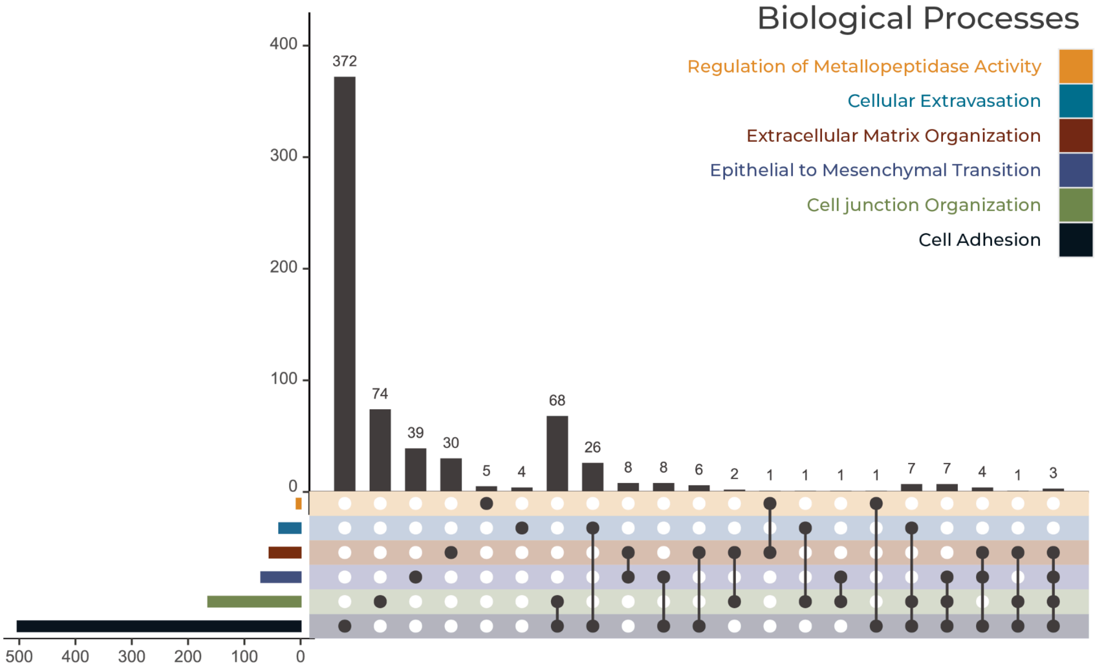
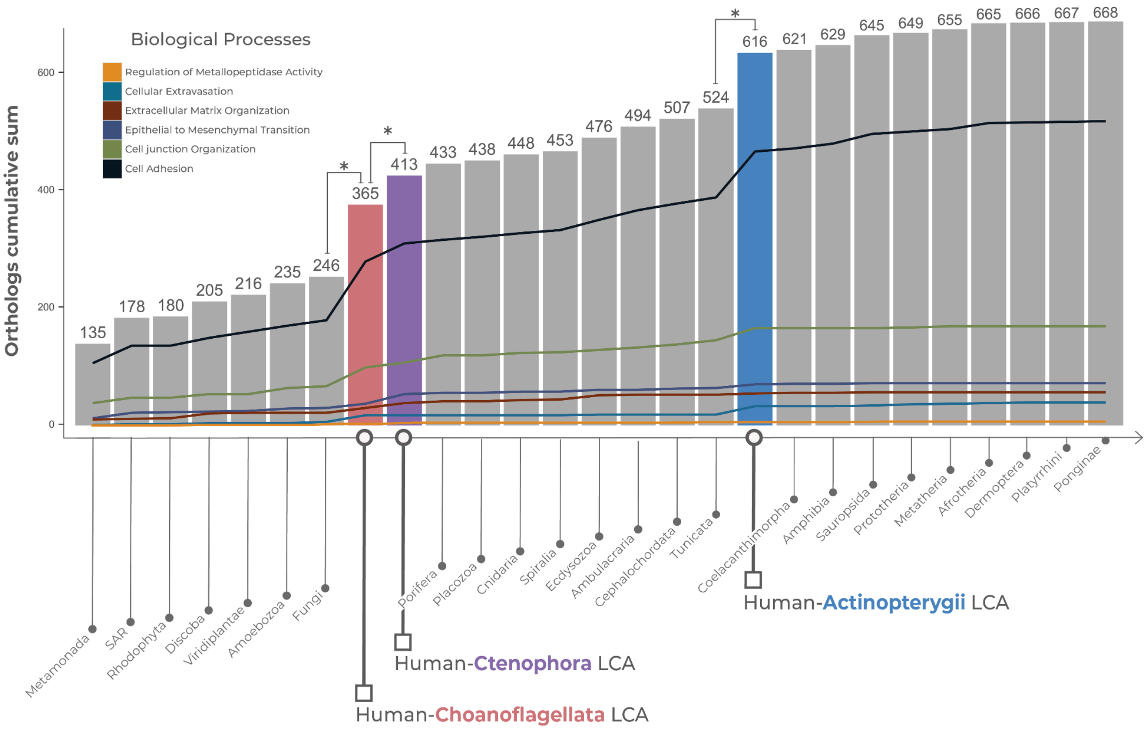


Figure 2: Genes involved in cellular extravasation are more commonly shared across multiple biological processes than exclusive to this pathway. This plot illustrates the distribution of genes across six biological processes associated with the early stages of metastasis. The intersections highlight the extent to which genes are shared among multiple processes.

Our evolutionary rooting analysis revealed three major clades of significant orthologs emergence for metastasis-related processes. The clades corresponding to the last common ancestors of Human-Choanoflagellata, Human-Ctenophora, and Human-Actinopterygii showed a statistically significant overrepresentation of rooted OGs compared to other clades (modified z-test, q-value < 0.05; Figure 3; Supplementary Table 1).



Evolutionary origins of early metastasis-related genes are anchored in the emergence of multicellularity and mitosis. The bar chart displays the number of orthologs associated with early metastasis; each bar's height indicates the total count of orthologs that were rooted in that specific clade. These counts are broken down by contribution from the six biological processes, shown as line charts color-coded. Significant outliers in the count of rooted orthologs are highlighted in color and marked with an asterisk (modified Z-score > 3.5, $p < 0.05$).

3.2. Hypergeometric Test and Rooted Orthologs Distribution

To elucidate the evolutionary patterns of each biological process, we performed a hypergeometric test on the orthologs' rooting distribution to identify which processes were statistically enriched within each clade of the phylogenetic tree (Figure 4).

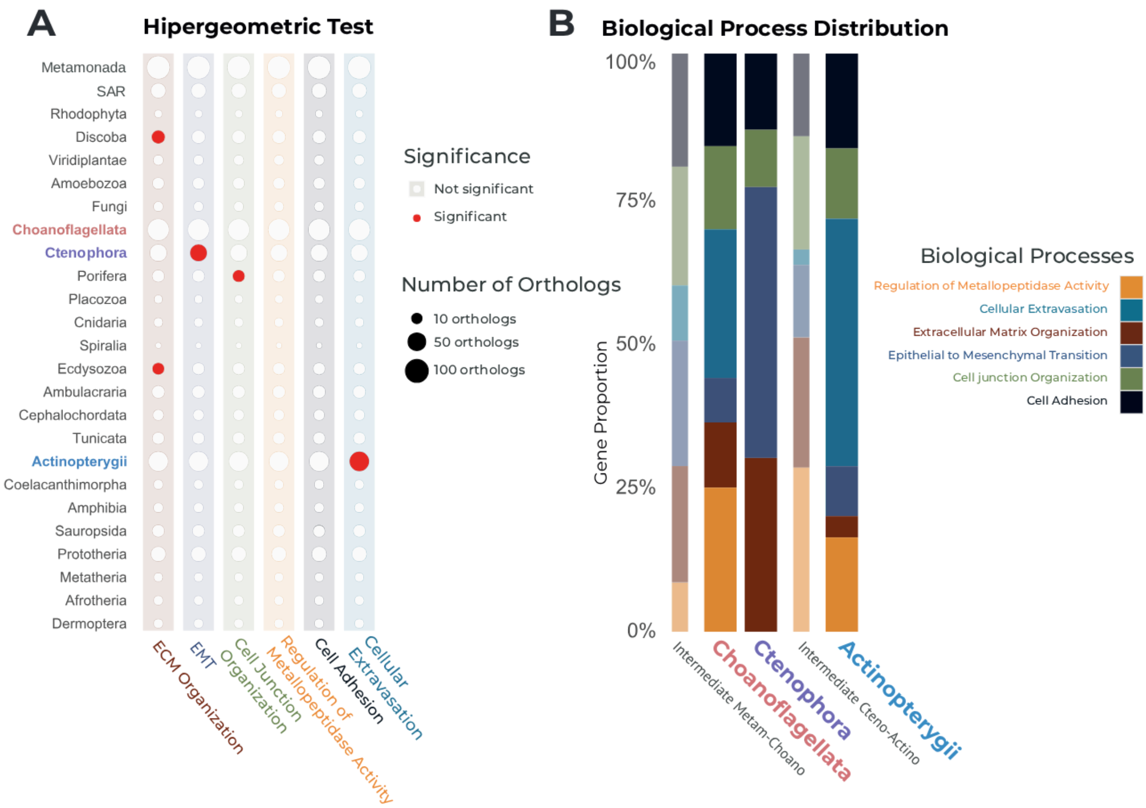


Figure 4: Divergent evolutionary origins - ECM components are enriched in unicellular ancestors, while other processes are enriched after the advent of multicellularity. (A) The enrichment of the six biological processes across each clade. Dot colors represent the significance (q-value < 0.05) from a hypergeometric test, indicating whether a process is significantly over-represented among the orthologs rooted in a given clade. **(B)** Distribution of orthologs origins for each of the six biological processes. The bar charts display the percentage of orthologs for each process that originated in a given clade. Thicker bars highlight the major clades of orthologs' emergence previously identified by the modified z-test (see Figure 3), while thinner bars denote intermediate clades.

The enrichment analysis of orthologs for key metastasis-related biological processes revealed distinct and significant enrichments at specific evolutionary junctures (Fig. 4). Our analysis showed no significant enrichment for the queried processes in the earliest-diverging clades, such as the Human-Metamonada LCA. The first significant enrichments were identified in pre-metazoan and early-metazoan lineages. Specifically, the ECM organization was significantly enriched in the Discoba clade (q-value = 0.024). Following this, at the base of the Metazoa, we observed a significant enrichment for EMT in Ctenophora (q-value = 0.001) and for cell junction organization in Porifera (q-value = 0.026). The process of ECM organization was again significantly enriched later in the bilaterian lineage Ecdysozoa (q-value = 0.049). Finally, within the vertebrate radiation, a significant enrichment for cellular extravasation was detected in Actinopterygii (q-value = 0.016). In contrast, processes such as cell adhesion and the regulation of metallopeptidase activity did not show a statistically significant enrichment in any single clade analyzed.

4. Discussion

Metastasis does not represent a *de novo* evolutionary innovation but rather the co-option of deeply conserved developmental and homeostatic programs. Our analysis indicates that the molecular basis for cancer dissemination was not a late vertebrate invention but was assembled in modular stages over a deep evolutionary timescale. The biological processes that underpin early stages of the metastatic cascade—cell adhesion, ECM organization, EMT, regulation of metalloprotease activity, extravasation, and cell junction organization—are deeply rooted in eukaryotic history, with key elements emerging before the Cambrian explosion and the origin of Metazoa. This assembly reflects the redeployment of a pleiotropic ancestral functional module into specialized functional modules. This evolutionary trajectory suggests a gradual assembly of an ancestral functional module later redeployed for tumor invasion and survival.

Our findings show that genes implicated in the initial phases of metastasis often serve multiple biological functions rather than belonging exclusively to a single pathway. This reflects pleiotropy, a core principle of evolutionary genetics in which one gene influences diverse traits (Fares et. al. 2020; Kiri et al. 2024). While pleiotropy provides robustness and efficiency in normal physiology, it creates vulnerabilities that cancer cells can exploit.

Functional module consolidation appears to have emerged within unicellular lineages. We identified significant enrichment of orthologs for ECM organization in the Human-Discoba LCA, indicating that a coordinated ECM functional module predates Metazoa. This is supported by genomic studies of discoba lineage that build functional ECMs during aggregative multicellularity, with genes for ECM construction, signaling, and motility (Sheikh et. al., 2024).

In contrast, holozoans such as choanoflagellates possess certain individual ECM domains, but these domains are not organized into the characteristic architectures of metazoan ECM proteins, and several other ECM domains are absent (King et. al., 2008; Hynes, 2012). The enrichment of ECM organization-related orthologs in Human-Discoba LCA suggests that this machinery evolved independently and earlier than once assumed. This may represent pre-adaptation for complex multicellularity, which itself has evolved multiple times across the eukaryotic tree (Hynes, 2012; Parfrey and Lahr, 2013; Draper et. al., 2019). In cancer biology, the ECM provides the structural and biochemical scaffold on which tissues are built, and its dynamic remodeling is a prerequisite for the initial stages of tumor invasion.

The transition to animal multicellularity required both tissue cohesion and cellular plasticity. Our results suggest that these opposing functions were established in the earliest-branching animal phyla. We observed a significant enrichment for cell junction organization in Human–Porifera LCA, which challenges the classical view of sponges as lacking true epithelia. Molecular evidence demonstrates that sponges express orthologs of key junctional scaffold proteins, such as MAGI, as well as transmembrane receptors like tetraspanins, within epithelial-like cell layers, indicating an early molecular basis for epithelial integrity (Adell et. al., 2004).

We also found enrichment for EMT-associated orthologs in the Human–Ctenophora LCA. EMT is a fundamental developmental process involving the loss of epithelial characteristics and the gain of a migratory, mesenchymal phenotype; it is also a critical driver of metastatic progression (Lamouille et. al., 2014). The ctenophore genome has been described as a "physiological oncogenome" and exhibits a constitutively active Wnt/β-catenin pathway due to the physiological absence of key negative regulators like Dickkopf proteins (Sinkovics et. al., 2015; Ye et. al., 2024). As the Wnt pathway is a primary inducer of EMT, this suggests an ancient link between developmental signaling, epithelial plasticity, and pathways that are later hijacked during metastasis.

Together, the presence of cell junction machinery in Human-Porifera LCA and EMT machinery in Human-Ctenophora LCA suggests that the balance between epithelial stability and plasticity was established at the dawn of animal evolution. Metastasis can thus be seen as the dysregulation of this ancient equilibrium.

Further enrichment of ECM organization orthologs in the Human–Ecdysozoa LCA aligns with the conserved “basement membrane functional module” shared across bilaterians (Hynes and Zhao, 2000; Whittaker and Hynes, 2002; Huxley-Jones et. al., 2007). This core set of proteins, including type IV collagen and laminins, is essential for tissue organization across protostomes and deuterostomes (Hynes, 2012). Its presence in cnidarians and even placozoans underscores its early origin and long-term role as a structural foundation for metastasis.

The later stages of the metastatic cascade, particularly active invasion and dissemination, appear to be vertebrate innovations. We identified enrichment of cellular extravasation-related orthologs in Actinopterygii. Extravasation, required for both immune function and metastatic spread, depends on vertebrate-specific systems such as a high-pressure closed vasculature and adaptive immunity. Many molecules involved in leukocyte trafficking are co-opted by cancer cells (Feuermann et. al., 2025; Panara et. al., 2025).

The machinery for cellular motility, however, is far older. Filopodia formation, essential for migration and extravasation, is based on actin-regulating proteins conserved in early eukaryotes such as Human-Discoba LCA (Cavalier-Smith and Chao, 2003; Sebé-Pedrós et. al., 2013; Arthur et. al., 2021). This reflects a recurring evolutionary theme: ancient modules repurposed for increasingly complex functions. Vertebrate extravasation thus represents the adaptation of older components into a new context, later hijacked by metastatic cells.

The analyzed processes exhibit distinct evolutionary dynamics across clades. The machinery for the EMT functional module, for example, shows an expansion in Ctenophora, consistent with stabilizing selection after the establishment of robust morphogenetic systems (Ozbek et al., 2010). In contrast, a later expansion occurred in orthologs associated with cellular extravasation during vertebrate evolution, likely driven by the increasing complexity of circulatory and immune systems. This trajectory resulted in a progressively larger repertoire of genes that could be subsequently co-opted during neoplastic progression.

In summary, our phylogenetic analysis suggests that metastasis does not represent a novel cancer-specific mechanism but a pathological redirection of ancient biological programs. Sequential acquisition of ECM organization, epithelial regulation, and later vertebrate-specific invasion machinery demonstrates that metastasis builds upon—and corrupts—the core processes of adhesion, communication, and development that enabled multicellular life.

5. Conclusion

This study highlights that metastasis should not be regarded as a novel invention of malignant cells, but as a pathological reconfiguration of pre-existing, deeply conserved biological mechanisms. By tracing phyletic patterns, we show that the components enabling cancer dissemination were gradually assembled across evolutionary history, with distinct functional modules arising at different clades. Rather than emerging abruptly in vertebrates, the metastatic toolkit reflects a layered evolutionary construction, later subverted to support tumor invasion, survival, and spread.

We have demonstrated a trajectory of gradual acquisition, beginning with the establishment of an ECM organization module in pre-metazoan eukaryotes (Human-Discoba LCA), which provided the foundation for tissue architecture. With the origin of Metazoa, a fundamental balance emerged between epithelial integrity, through the enrichment of biological processes for cell junction organization (Human-Porifera LCA), and epithelial plasticity, with the enrichment of machinery for EMT (Human-Ctenophora LCA). Finally, with the advent of vertebrate lineages, such as Actinopterygii, this ancestral functional module was refined to include specialized mechanisms for systemic dissemination, such as cellular extravasation.

These findings confirm that the biological complexity of metastasis results from the rearrangement of a pleiotropic ancestral functional module, assembled through successive evolutionary events. This evolutionary perspective redefines our understanding of metastasis, not as a set of unique cancer-specific traits, but as the pathological corruption of the very processes of adhesion, communication, and development that enabled the emergence of multicellular life.

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7. Declarations

7.1. Data Availability

All data supporting the findings of this study are included within the article and its supplementary materials. The code used for the analyses is available at GitHub (https://github.com/dalmolingroup/metastasisevolution).

7.2. Ethical Issues

This study was based exclusively on computational and publicly available genomic datasets. No experiments involving live vertebrates or human subjects were conducted.

7.3. Conflict of Interest

The authors declare that no conflict of interest could be perceived as prejudicial to the impartiality of the reported research.

7.4. Authors Contributions

G.M.A. conducted the investigation, performed formal analyses, and wrote the original draft. E.F. and R.S.F.D contributed to the conceptualization and validation of the study. They participated in writing, reviewing, and editing, providing their expertise in cancer biology and contributing to selecting biological processes associated with the early stages of metastasis. G.M.A., J.V.F.C. and B.W. provided resources, software, and methodological support, including assistance with the computational environment and implementation of the GeneBridge tool. R.J.S.D. supervised the study, contributed to conceptualization, and oversaw project administration, while D.M.C. acted as co-supervisor and contributed to data curation, formal analysis, and interpretation of results. All authors read and approved the final version of the manuscript.

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